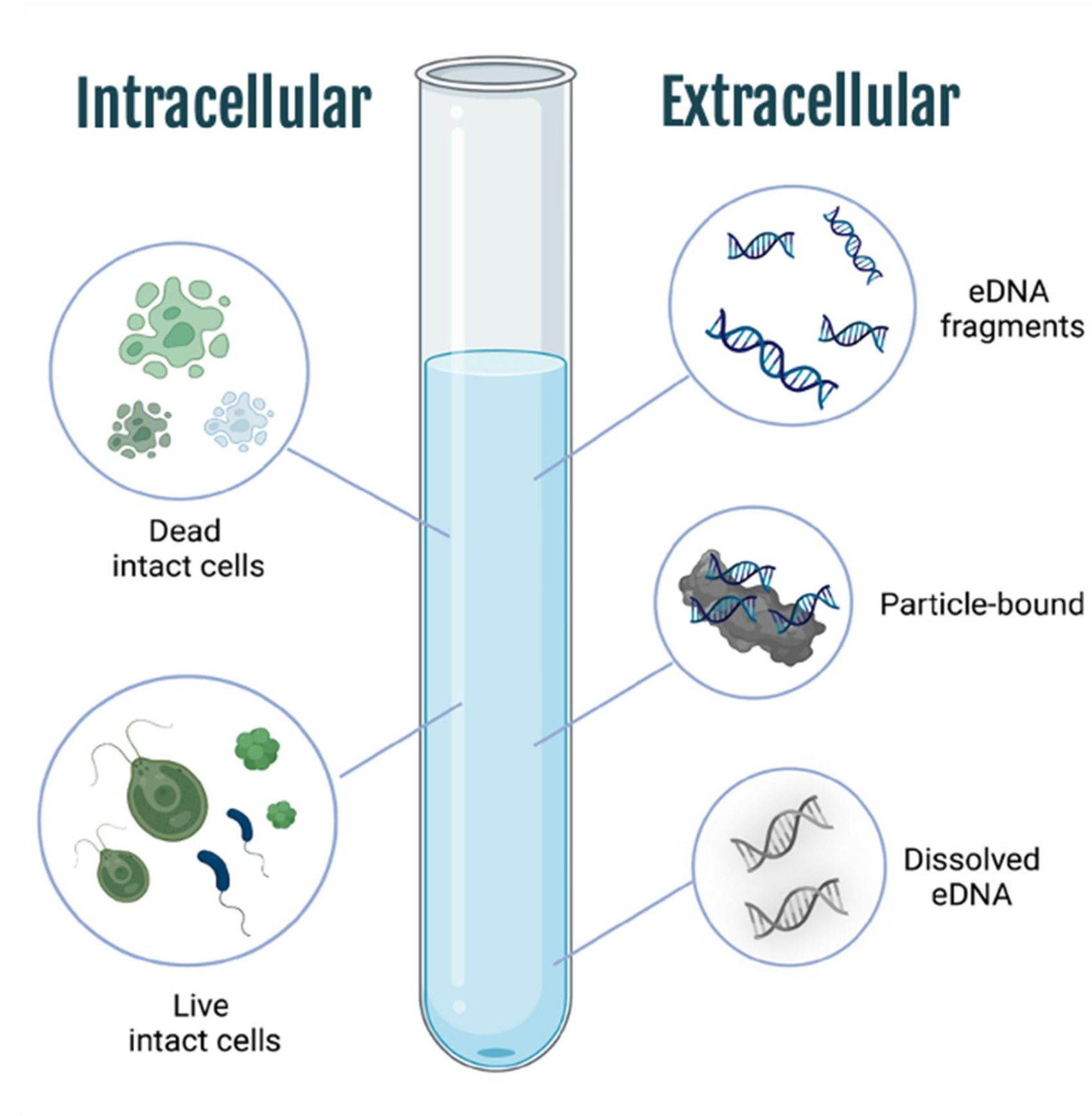
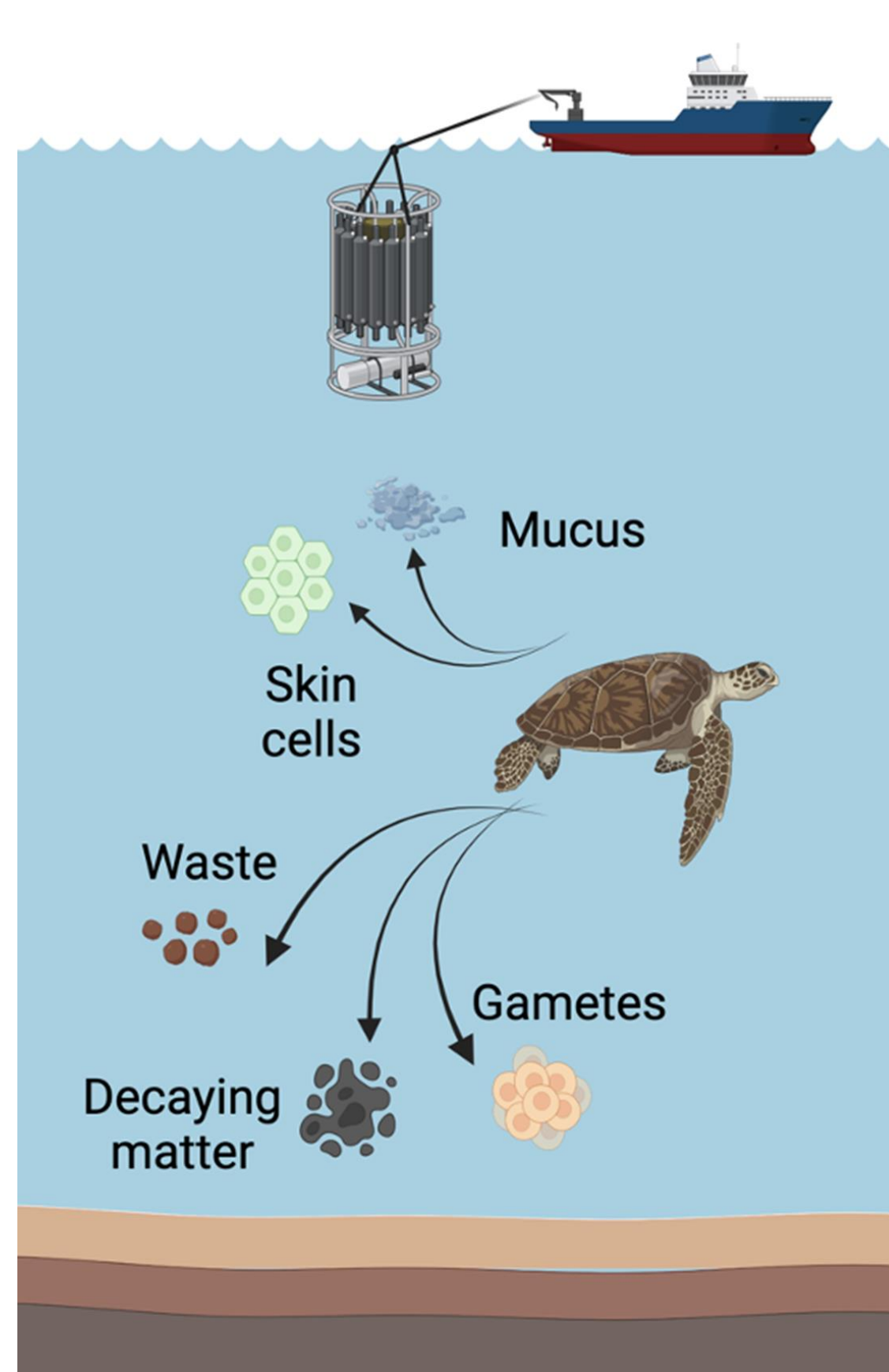


Biodiversity observation ain't easy

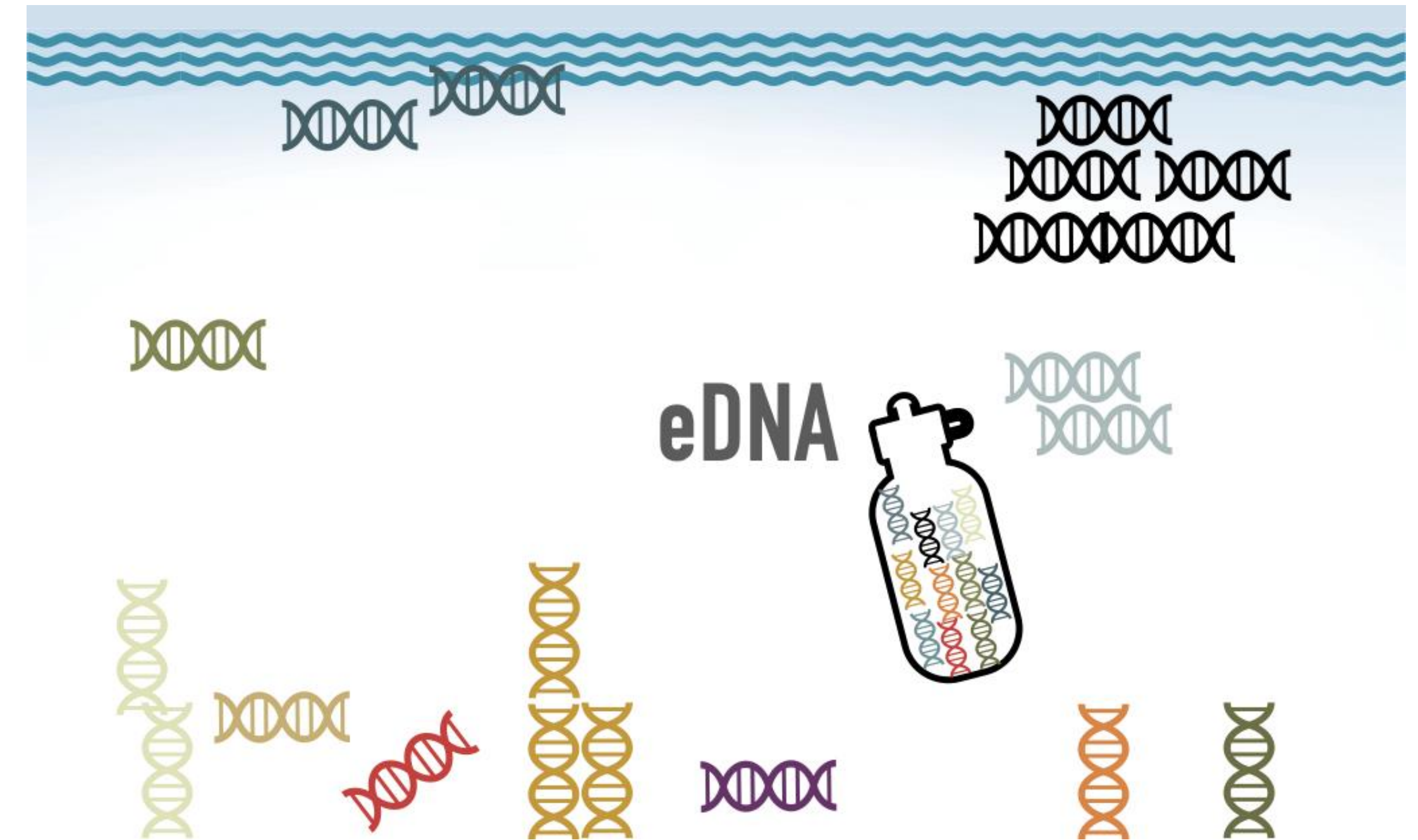
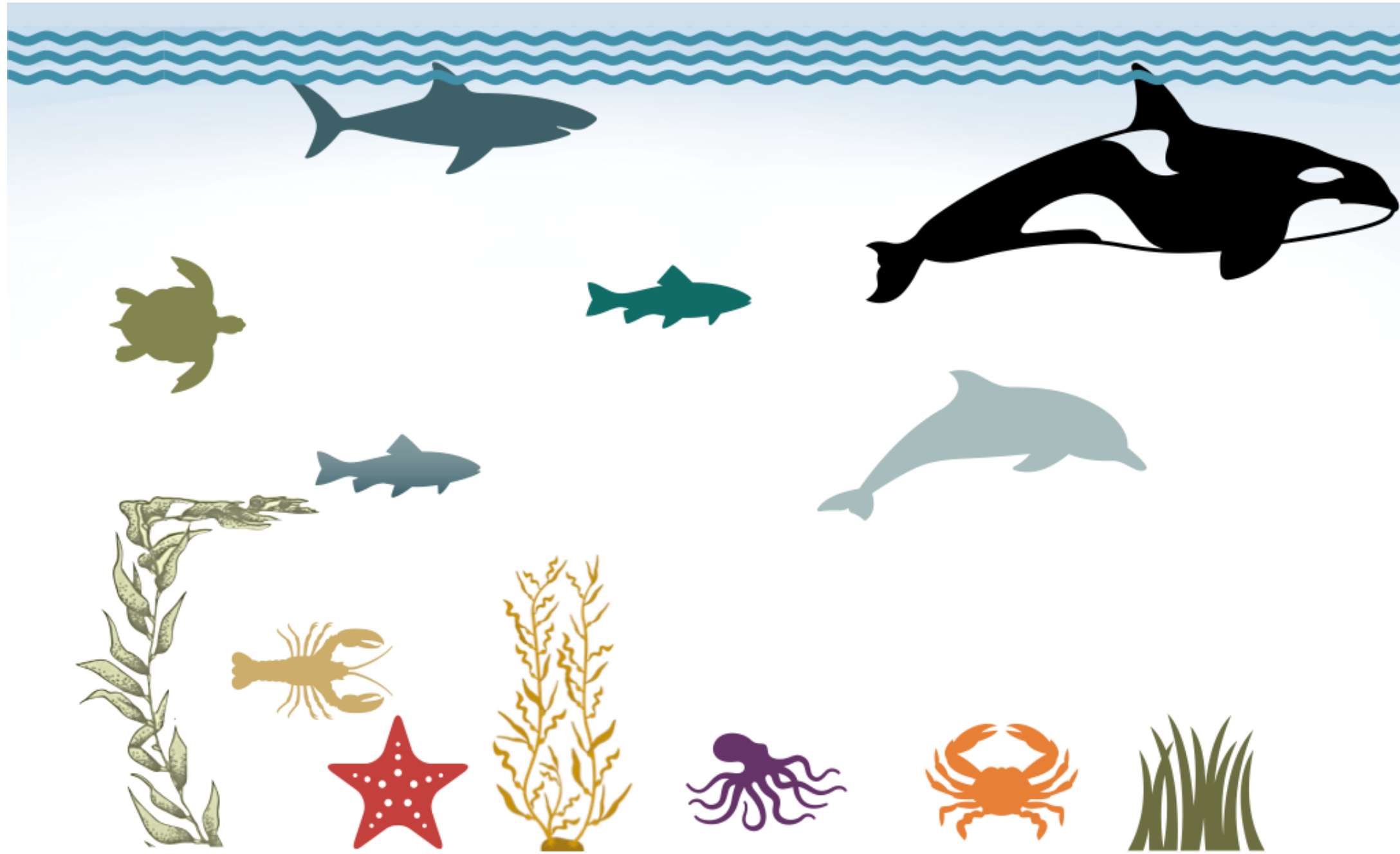
- Time and Labor intensive
- Expensive
- Requires taxonomic expertise
- Dominated by conspicuous or commercial species
- Conditions dependent

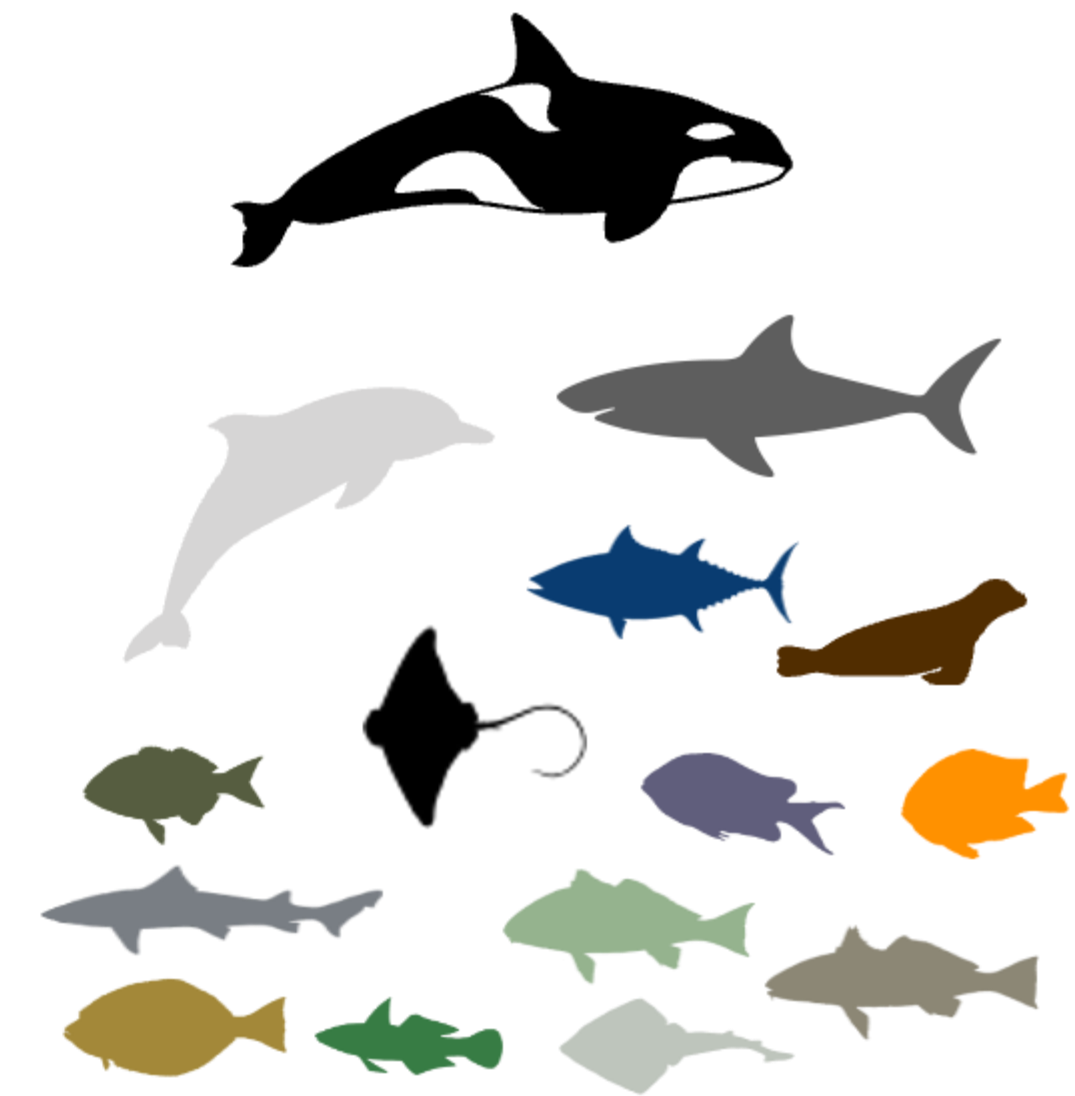
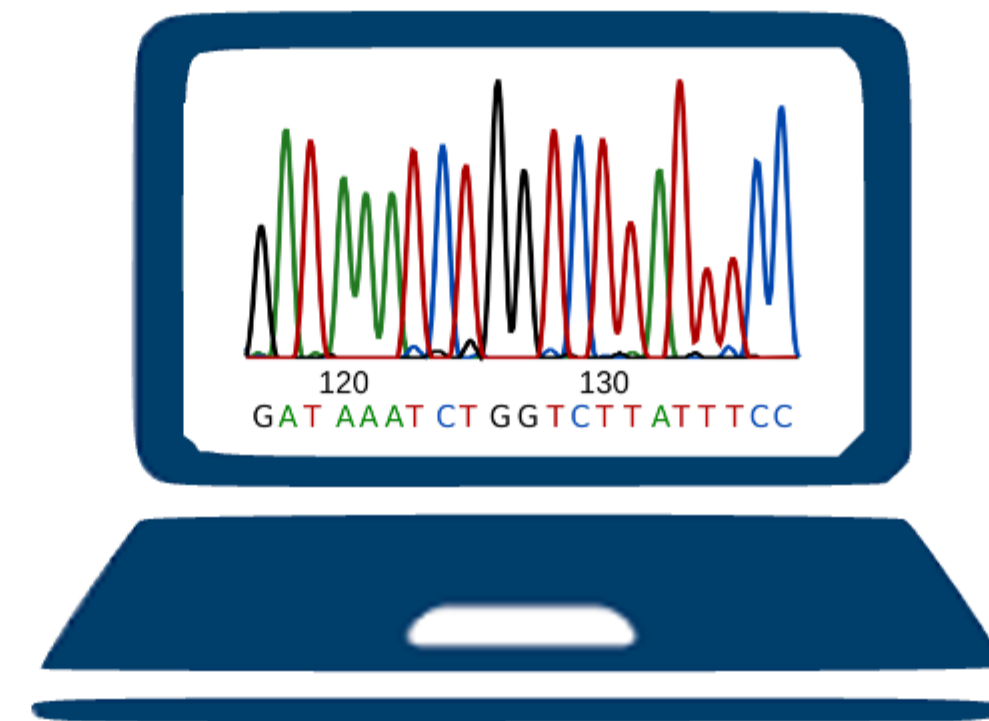
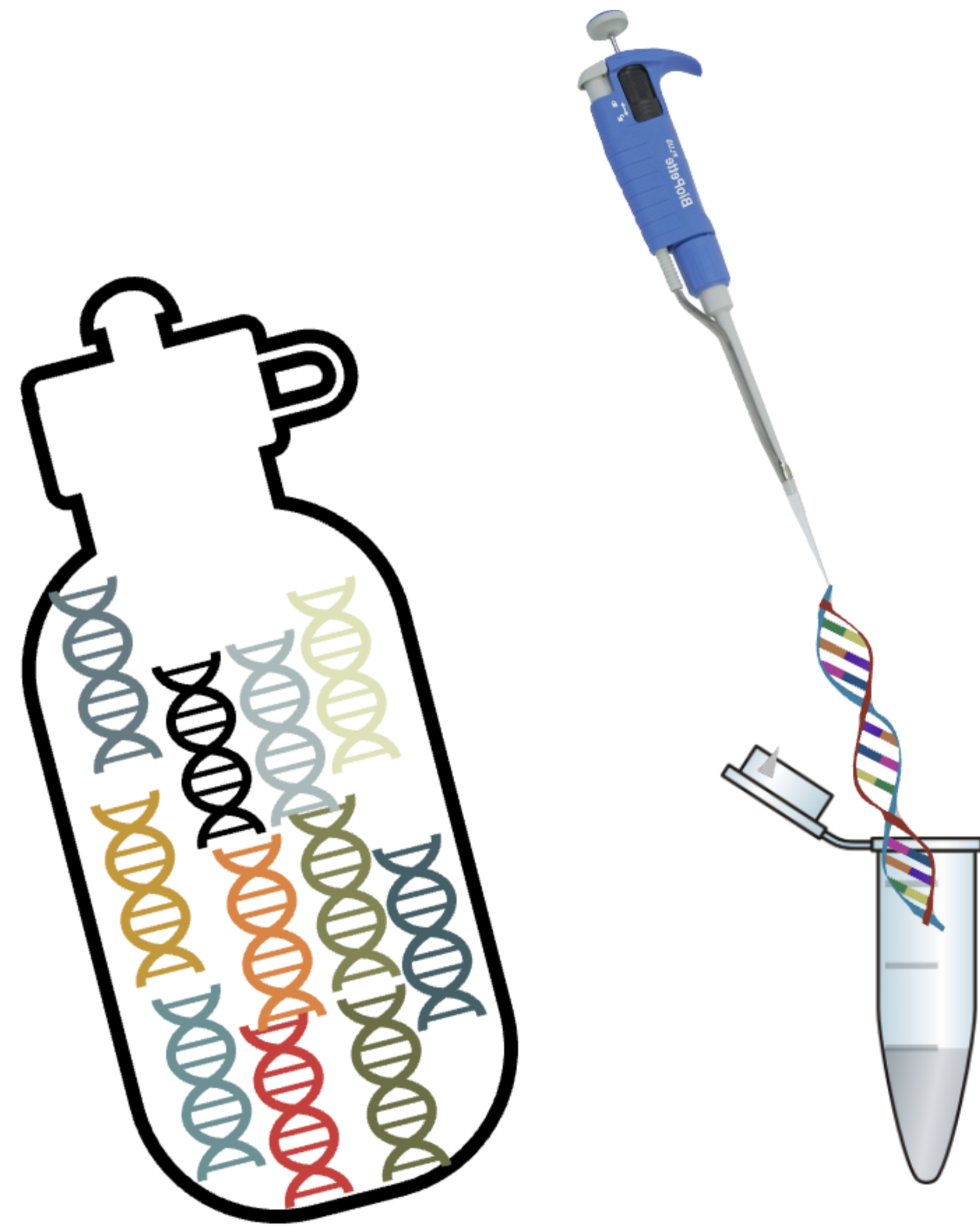


There are many types of evidence of biodiversity.



eDNA can reduce the resources needed to observe biodiversity





Sample → **Process** → **Sequence** → **Reconstruct**



Strengths of eDNA



Sampling

Ease of sampling, collect samples remotely

Minimal environmental impact

Filters, DNA extracts, and sequences can be preserved long term, allowing reanalysis



High Sensitivity

Improved detection of rare and cryptic species

Detect broad diversity of species simultaneously, from microbes to mammals



Standardization of biases

Ability to standardize taxonomic analyses

Minimize biases and increase comparability of data across sampling programs



Burden

Low cost per sample

Fast data turnaround times



Declining Cost of Sequencing

2001

\$10,000,000.00

Per DNA Gigabase

Declining Cost of Sequencing

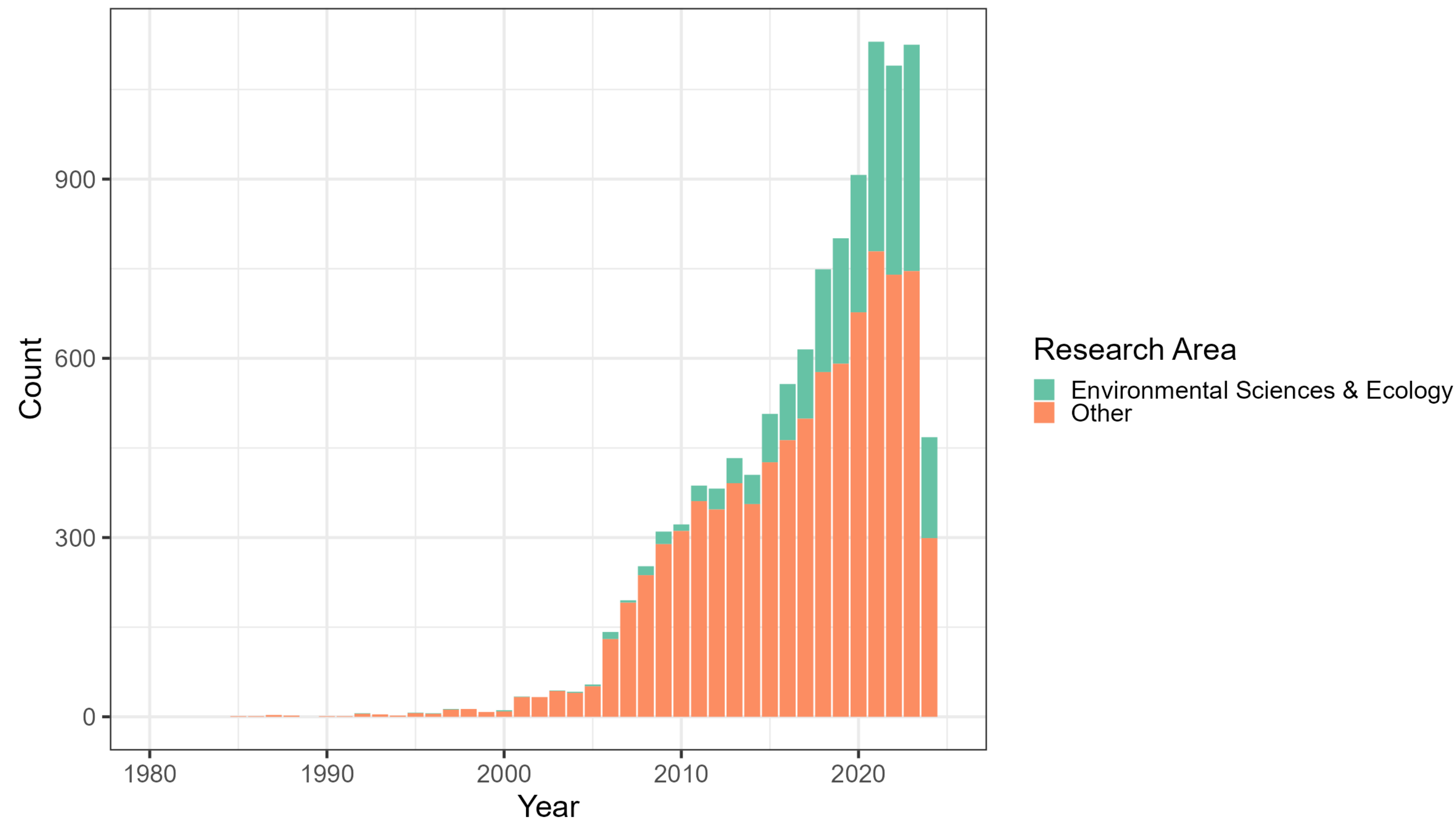
2022

\$2.00

Per DNA Gigabase

Abundance of eDNA Publications Over Time

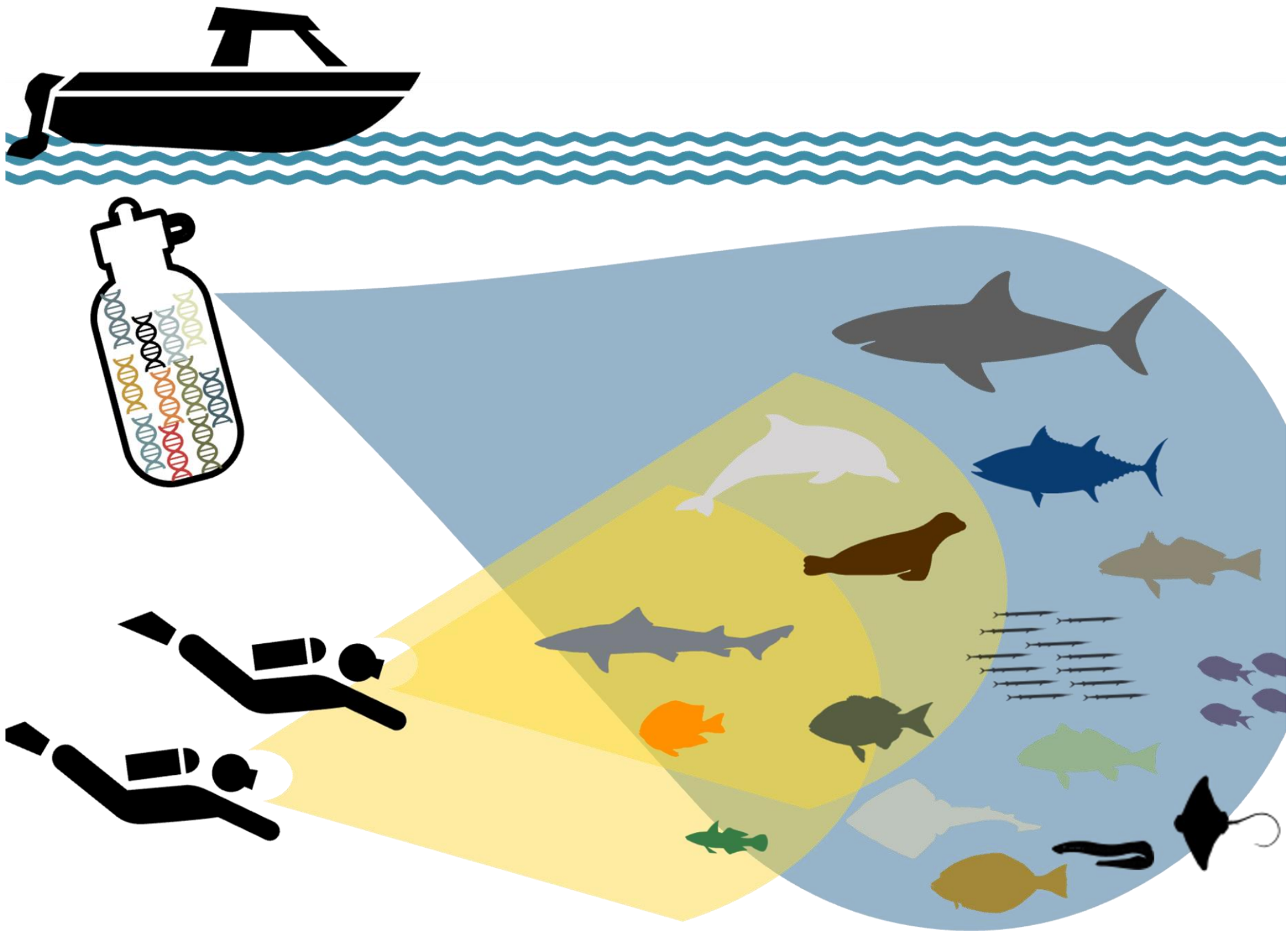
based on Chavez et al. 2021, Figure 3



Publications from a search of the Web of Science within environmental domains papers using the words “environmental DNA” or “eDNA.” Publications have increased exponentially in recent years, demonstrating the growing popularity of this method. Colors represent Web of Science 'Research Areas', with 'Other' being the aggregation of everything that is not 'Environmental Sciences and Ecology.'



eDNA helps, but it's not a silver bullet



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